

PM Surya Ghar: Muft Bijli Yojana (PM Ghar Bijli Yojana) — Complete Detailed Guide

1. What is PM Surya Ghar: Muft Bijli Yojana?

PM Surya Ghar: Muft Bijli Yojana is a Government of India rooftop solar scheme launched on **13 February 2024** to promote solar energy adoption in residential houses.

The main objective:

- Install rooftop solar systems in **1 crore Indian households**
- Help families reduce electricity bills
- Provide financial support through solar subsidy
- Increase clean renewable energy usage

The scheme is managed through the Ministry of New and Renewable Energy (MNRE), Government of India.

2. How Does the Scheme Work?

The basic process:

Your House Roof



Solar panels installed



Solar panels generate electricity



Electricity is used in your home



Extra electricity goes to the grid through net metering



You save money on electricity bills

3. Solar System Types Under the Scheme

A) On-Grid Solar (Most Common)

This is the system generally used under PM Surya Ghar.

Components:

- Solar panels
- Solar inverter
- Net meter
- Mounting structure

- DC/AC cables

Working:

Daytime:

- Solar produces electricity
- Home appliances use solar power

Extra power:

- Sent to electricity grid

Night:

- Electricity comes from grid

Advantages:

- ✓ Lower cost
- ✓ Subsidy available
- ✓ No battery maintenance

Disadvantage:

- ✗ Usually does not work during power cuts
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B) Hybrid Solar System

Includes:

- Solar panels
- Inverter
- Battery storage

Advantages:

- ✓ Works during power cuts
- ✓ Stores extra electricity

Disadvantages:

- ✗ Higher cost
 - ✗ Battery replacement needed after some years
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4. Subsidy Details

Government subsidy depends on solar capacity.

Solar Capacity Approx Subsidy

1 kW	₹30,000
2 kW	₹60,000
3 kW and above	Up to ₹78,000

Example:

3 kW Solar System

Approx installation cost:
₹1.5 lakh – ₹2 lakh (varies)

Government subsidy:
₹78,000

Your approximate payment:
₹70,000–₹1.2 lakh

(Actual price depends on vendor, equipment, location, and installation.)

5. Electricity Generation Estimate

Solar generation depends on sunlight.

Average Indian conditions:

1 kW system

Produces:
≈ 100–120 units/month

Yearly:
≈ 1200–1400 units

2 kW system

Produces:
≈ 200–240 units/month

Yearly:
≈ 2400–2800 units

3 kW system

Produces:
≈ 300–360 units/month

Yearly:
≈ 3600–4200 units

6. How Many Panels Are Needed?

Modern solar panels:

One panel:
≈ 540W–600W

Approx:

1 kW

2 panels

2 kW

4 panels

3 kW

5–6 panels

Roof requirement:

Approx:

- 1 kW → 80–100 sq ft
 - 3 kW → 250–300 sq ft
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7. Eligibility Criteria

You can apply if:

- ✓ Indian citizen
 - ✓ Residential electricity connection
 - ✓ Own/occupy a suitable house
 - ✓ Roof available for solar installation
 - ✓ Bank account available for subsidy transfer
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8. Required Documents

Keep these ready:

Identity

- Aadhaar card
- PAN card (if required)

Electricity

- Latest electricity bill
- Consumer number

Bank

- Bank account details
- Cancelled cheque/passbook

Communication

- Mobile number
 - Email ID
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9. Application Process Step-by-Step

Step 1: Registration

Visit:

[PM Surya Ghar Official Portal](#)

Enter:

- State
 - Electricity distribution company
 - Electricity consumer number
 - Mobile number
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Step 2: Submit Application

Fill:

- Personal details
 - Address
 - Solar capacity requirement
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Step 3: DISCOM Approval

Your electricity company checks:

- Connection details
- Technical feasibility

After approval, you can proceed.

Step 4: Select Solar Vendor

Choose an approved vendor from the portal.

Vendor installs:

- Panels
 - Inverter
 - Wiring
 - Structure
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Step 5: Inspection

Electricity company checks:

- Installation quality
 - Safety
 - System specifications
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Step 6: Net Meter Installation

A bidirectional meter is installed.

It measures:

- Electricity imported from grid
 - Electricity exported to grid
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Step 7: Subsidy Payment

After completion:

- Submit bank details
 - Government transfers subsidy directly
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10. Example: Ahmedabad/Gujarat Household

Suppose:

Monthly electricity use:
300 units

You install:
3 kW rooftop solar

Solar generation:
≈ 300+ units/month

Possible result:

Before:
Electricity bill = ₹2000–₹3000/month

After solar:
Bill can reduce significantly (depending on usage, tariff, and generation)

11. Loan Facility

Many banks provide loans for rooftop solar installation.

Possible benefits:

- Lower upfront payment
 - Pay through EMI
 - Subsidy reduces effective cost
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12. Maintenance

Solar panels require:

Cleaning:

- Every few weeks/months depending on dust

Maintenance:

- Very low

Life:

Panels:

20–25 years

Inverter:

Usually around 5–10 years (varies by model)

13. Common Mistakes to Avoid

- ✗ Choosing unauthorized vendors
 - ✗ Installing without DISCOM approval
 - ✗ Buying very cheap low-quality panels
 - ✗ Ignoring roof shade
 - ✗ Not checking warranty
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14. Advantages for Homeowners

Financial

- Lower electricity bills
- Long-term savings

Environmental

- Less carbon emissions
- Clean energy

Property

- Adds value to property

Benefits of PM Surya Ghar: Muft Bijli Yojana

1. Electricity Bill Reduction

- You can reduce your monthly electricity bill by generating your own solar power.
- The scheme targets providing **up to 300 units of electricity per month** through rooftop solar benefits for eligible households.

Example:

- Before solar: ₹2000/month electricity bill
 - After solar: Bill can become very low (depends on your usage and solar generation)
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2. Government Subsidy (Big Benefit)

Government gives financial assistance:

- **1 kW solar** → ₹30,000 subsidy
- **2 kW solar** → ₹60,000 subsidy
- **3 kW or above** → up to ₹78,000 subsidy

The subsidy is provided through the scheme after installation and approvals.

Example:

3 kW solar system cost:
≈ ₹1.5–2 lakh

Government support:
₹78,000

Your remaining cost becomes much lower.

3. Long-Term Savings

Solar panels usually work for around **20–25 years**.

Benefits:

- One-time installation
 - Years of electricity savings
 - Less dependence on increasing electricity prices
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4. Generate Your Own Electricity

You become a small electricity producer:



Sunlight



Solar panels



Electricity for your home

This gives more energy independence.

5. ✂ Extra Electricity Benefit (Net Metering)

If your solar system produces extra electricity:

- Extra power can go to the electricity grid
 - It can be adjusted through net metering rules, reducing your bill further.
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6. Environmental Benefit

Solar energy helps:

✓ Reduce carbon emissions

- ✓ Reduce fossil fuel dependency
 - ✓ Promote clean energy
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7. Loan Support

Eligible households can access easier financing options for rooftop solar installation, reducing the need for a large upfront payment.

8. Property Value Benefit

A house with rooftop solar can become more attractive because:

- Lower electricity expenses
 - Renewable energy setup already installed
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9. Best For Families With:

- ✓ High electricity bills
 - ✓ Independent house
 - ✓ Good sunlight on roof
 - ✓ Long-term house ownership
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15. Frequently Asked Questions

Q. Is electricity completely free forever?

No. The scheme supports solar generation. Your savings depend on your solar production and electricity usage.

Q. Can tenants apply?

Usually the electricity connection holder/eligible consumer applies.

Q. Can I install 5 kW or 10 kW?

Yes, but subsidy is limited according to scheme rules.

Q. Does government install panels?

No. Approved vendors install the system.

Q. Can I sell electricity?

Usually excess electricity is adjusted through net metering rules, not a direct business sale.

